SUPPLEMENTARY MATERIAL

Critical Overflow in Weakly Coupled Two-Choice Allocation

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This supplement develops the auxiliary estimates, numerical certificates, statistical methods, and sensitivity experiments used in the main article.

SM1 The centered Gini inequality

For a finite zero-mean real law Y,

$$\mathbb{E}|\sinh Y - \sinh Y'| \geq \mathbb{E}|\sinh Y| \geq \mathbb{E}\cosh Y - 1.$$

The second inequality holds pointwise. For the first, decompose a zero-mean law into zero-mean two-point laws across zero, plus any atom at zero, by transporting the positive first-moment mass against the negative first-moment mass. For a component on $a, -b$, $a, b > 0$, the difference between the two sides is

$$\frac{(a-b)(b\sinh a - a\sinh b)}{(a+b)^2} \geq 0,$$

since $\sinh(t)/t$ increases on $(0,\infty)$. The Gini functional of a real law is concave, as seen from $2\int F(t)(1-F(t))\,\mathrm{d}t$, whereas the absolute first moment is linear. Mixing the components proves the result. Truncation and mean correction extend it when the stated exponential moments are finite. Rank summation by parts then applies this inequality to suffix deficit $\delta t(1-t)$, giving the potential drift used in Sections 3 and 4.

SM2 Auxiliary process estimates

We derive the dispersion, empirical approximation, and tracking estimates used in the main proofs, followed by the label-forgetting coupling constants.

SM2.1 Dispersion bound

We prove main-article Lemma 3.1 with $\theta = 1/200$, $K_0 = 16/3$, and $I = (1-\varepsilon)/2$.

Proof. Order each community by increasing load, placing left buckets before right buckets at equal load. Use a fixed label order within each side. The resulting order agrees with the left tie rule. For the heaviest k buckets in this order put $t = k/q$. If their two side counts are a, b, their internal selection probability is at most $ab/m^2 \leq t^2$. Cross mass entering this suffix is $x \leq \varepsilon t$, while cross mass entering the community is $y \geq x$. Conditional on a community update, its suffix probability is at most

$$\frac{It^2 + \varepsilon t}{I + \varepsilon t} \leq t - \delta t(1-t), \qquad \delta = \frac{I-\varepsilon}{I+\varepsilon} = \frac{1-3\varepsilon}{1+\varepsilon} \geq \frac{1}{5}.$$

The global inclusion probability of a vertex is $1/q$, so its conditional update probability is at most $8/(3q)$. Rank summation by parts, for sorted $g_i = \sinh(\theta\xi_i)$, gives

$$\sum_i (p_i - q^{-1})g_i \leq -\frac{\delta}{2}\mathbb{E}|\sinh Y - \sinh Y'| \leq -\frac{\delta\Gamma_s}{2q} + \frac{\delta}{2}.$$

Here Y is uniform on the centered $\theta\xi_i$ and Y' is independent. The zero-mean inequality $\mathbb{E}|\sinh Y - \sinh Y'| \geq \mathbb{E}|\sinh Y| \geq \mathbb{E}\cosh Y - 1$ is proved in Section SM1. For the centered increment $u_i = \mathbf{1}_{i \text{ selected}} - 1/q$, its second moment is less than $3/q$. Taylor's formula bounds the total cosh remainder by $4\theta^2\Gamma_s/q$. Since $\theta \leq \delta/16$,

$$\mathbb{E}[\Delta\Gamma_s \mid C_s \text{ updates}] \leq -\frac{\theta\delta}{4q}\Gamma_s + \frac{\theta\delta}{2}.$$

Each community updates with probability at least $3/8$. The total drift is therefore at most

$$-\frac{3\theta\delta}{32q}\Gamma + \frac{\theta\delta}{2}.$$

The recurrence from $\Gamma_0 = 2q$ proves (3.1). □

SM2.2 Fixed-window empirical estimate

Use the translated initial state and constants in main-article section 5. The mean clock is $u = k/q$, and the fluid starts from the exact empirical profile. We prove the uniform estimate (5.1) for fixed B, R.

A valid explicit bound in (5.1) is

$$\eta_q(B,R,K_1) = e^{4B}\left[8(2K_1+1)\sqrt{B/q} + 4BC_{R,B}e^{-\theta(K_1-1)} + 8B/q\right],$$
$$C_{R,B} = 2e^{\theta}R\exp\{2B(e^{\theta}-1)\}.$$

Proof. Each event increments exactly one tail coordinate by $2/q$. Its conditional mean increment is exactly $F(X_k^q)/q$, where F is the field in (2.2). Subtract these conditional means to obtain a tail-valued martingale at the grid times. Each coordinate has expected quadratic variation at most $4B/q$. Doob's inequality bounds its expected maximum by $4\sqrt{B/q}$. There are $2(2K_1+1)$ retained coordinates, indexed by $|k| \le K_1$.

A bucket has selection probability at most $2/q$ at each internal event. Therefore the expected sum of its translated, uncentered cosh moments grows by at most the factor $1+2(e^{\theta}-1)/q$ per event. The initial sum of the two side moments is at most $e^{\theta}R$. The probability that the selected tail increment lies outside the retained coordinates is at most $C_{R,B}e^{-\theta(K_1-1)}$. The exponential moment and selection-probability bounds give this tail estimate. Summing actual outside increments and their nonnegative compensators gives expected outside martingale norm at most $4BC_{R,B}e^{-\theta(K_1-1)}$. Initial tails add no martingale error, since the fluid starts from the exact empirical profile.

The field is 4-Lipschitz in tail ℓ_1. A single interpolated empirical step has length at most $2/q$. Replacing the stepwise field integral by the interpolated one costs at most $8B/q$. Gronwall gives (5.1). □

SM2.3 Moments for profile tracking

The following calculation gives main-article (5.2).

For the real product process, let W_n be the sum of its four side Wasserstein distances to the corresponding orbit laws. Each community uses its actual mean $M_s(n)/q$. Let Ψ_n be the sum of the four empirical cosh moments centered at these means. Lemma 3.1 implies

$$\sup_n \mathbb{E}\Psi_n \le C_E = 32/3.$$

If J_n is the sum of the four empirical mean absolute centered deviations, Jensen gives

$$e^{\theta J_n/4} \le \Psi_n/2, \qquad \mathbb{E}J_n^2 \le 16C_E/\theta^2.$$

The second inequality uses $x^2 \le 2e^x$ for $x \ge 0$. The four orbit absolute deviations sum to at most $8/\theta$, since $V_\theta(\rho^u) \le 4$. Thus

$$\sup_n \mathbb{E}W_n^2 \le C_*^2 = (32C_E + 128)/\theta^2. \tag{SM2.1}$$

SM2.4 Label forgetting

This proves main-article Lemma 3.2, using its constants and notation.

Proof. Write P for the fixed transition kernel and $\mu = \mu_{n-L}$. Split

$$\mathrm{TV}(P^L\mu, \tau P^L\mu) \le \mathrm{TV}(P^L\mu, P^L\tau\mu) + \mathrm{TV}(P^L\tau\mu, \tau P^L\mu).$$

These two terms need different couplings. For the first, start A from μ and A' from τA. Map internal candidates by τ until the two tagged loads in A meet. This is legal uniform internal sampling and maintains $A' = \tau A$. Use the same actual cross edge; the relation can fail only when a cross touches a tag, with window probability at most $2\varepsilon\ell$. Once the tag loads meet the configurations agree, so common candidates keep them equal.

Before meeting or a bad cross, the higher tag has no larger internal update rate: a left tag at h has rate $(1-\varepsilon)y_h/q$, a right tag has rate $(1-\varepsilon)x_{h+1}/q$. The embedded positive gap is consequently dominated by a fair nearest-neighbor walk absorbed at zero. Its survival for J jumps from gap d is at most $\min(1, 4d/\sqrt{J+1})$, by reflection and the central binomial bound. Fresh direction coins give this domination in the adapted environment.

Each external uniform tag has expected centered cosh at most $2K_0$, hence expected initial gap at most $4K_0/\theta$. The community receives at least $L/4$ internal arrivals except with probability $e^{-L/48}$, by the binomial lower tail since $I \ge 3/8$. At the two window endpoints, all four tag deviations are at most $\ell/16$ except with probability $16K_0e^{-\theta\ell/16}$. Off these events, each tag gains at least $\ell/8$, since the community mean gains at least $\ell/4$. Put $K = \lceil \ell/4 \rceil$. If the tags have not met and no cross touches them, at least K embedded internal jumps have occurred. The nonmeeting event is therefore contained in the union of the stated bad events and survival of the dominating fair walk for the fixed number K of jumps. Reflection and $\mathbb{E}d \le 4K_0/\theta$ bound this last probability by $32K_0/(\theta\sqrt{\ell})$. This controls the first TV term with an additional $2\varepsilon\ell$.

For the second term compare the fixed kernel with its τ-conjugate, from the same law $\tau\mu$. The internal laws agree; the two swapped cross columns have total conditional mass $2/m$. Coupling all other mass gives another window cost $2\varepsilon\ell$. Adding the costs proves (3.2). □

SM3 Auxiliary numerical estimates

SM3.1 Analytic tails for the true orbit

We derive analytic tail bounds for the infinite periodic orbit used in the a posteriori certificate. Let $D_{LR}(u) = \mathbb{E}|L_u - R_u|$ with independent draws. Exact differentiation of the centered second-moment sum gives

$$\frac{\mathrm{d}}{\mathrm{d}u}\{\mathbb{E}(L_u - u)^2 + \mathbb{E}(R_u - u)^2\} = 2 - 2D_{LR}(u).$$

It follows by periodicity that $\int_0^1 D_{LR}(u)\,\mathrm{d}u = 1$. The function D_{LR} is 2-Lipschitz by the monotone marginal coupling, so its maximum is at most $3/2$ on the unit circle. Interlacing gives $0 \le \mathbb{E}L_u - \mathbb{E}R_u \le 1$. Jensen and positive/negative parts, at mean zero, then imply

$$\mathbb{E}(L_0)_+ \le 5/4, \quad \mathbb{E}(R_0)_+ \le 3/4, \quad \mathbb{E}(-L_0)_+ \le 3/4, \quad \mathbb{E}(-R_0)_+ \le 5/4.$$

For example $\mathbb{E}(L-R)_+ = (D_{LR} + \mathbb{E}L - \mathbb{E}R)/2$, while Jensen bounds the marginal positive part by that expression after accounting for the other marginal's sign of mean. The opposite-sign cases follow identically.

Integrating the front equation and matching the limit at $+\infty$ yields

$$F(r) = \int_{r-1}^{r} F(s-1)F(s)\,\mathrm{d}s, \qquad F(r) \le F(r-1)F(r-2). \tag{SM3.1}$$

The first-moment bounds give $F(2k) \le \min(1, 5/(4k))$ and $F(2k+1) \le \min(1, 3/(4k))$ for $k \ge 1$. Monotonicity and (SM3.1) give a rational superlinear right-tail recursion. Once $F(n) \le \eta < 1$, the uncomputed remainder obeys $F(r) \le \eta F(r-2)$ for $r \ge n+2$. Upward rounding and this geometric remainder account for the infinite tail even when finite arithmetic plateaus.

For the left anchor set $z_j = (x_j(0) + y_j(0))/2$ and $A_j = \int_0^1 x_j(u)y_j(u)\,\mathrm{d}u$. The pooled equation and the unit-period shift give $A_j = z_j$. Stochastic increase gives

$$z_j \le x_{j-1}(0)y_{j-1}(0) \le z_{j-1}^2.$$

If $z_{-2} \le s$, mean zero would require $3 \le \sum_{n\ge 0} s^{2^n}$. At $s = 89/100$ the latter sum is strictly below three: sum through $n = 5$ exactly and bound the rest by $s^{64}/(1-s^{64})$. Thus $G(-4) = 1 - F(-4) < 11/100$.

Write $a = 11/100$, $c = 1 - a/2 = 189/200$ and $\gamma = 18/25$. From (SM3.1), for $r \le -4$,

$$G(r) = \int_{r-1}^{r} [G(s) + (1 - G(s))G(s-1)]\,\mathrm{d}s \ge c\int_{r-2}^{r} G(s)\,\mathrm{d}s.$$

The kernel mass $c\int_0^2 e^{-\gamma s}\,\mathrm{d}s$ is at least one, as verified by a rational Taylor lower bound $e^{36/25} > 21/5$. Therefore $e^{-\gamma r}G(r)$ is superharmonic for the backward walk with normalized density proportional to $ce^{-\gamma s}$ on $[0,2]$. Starting at -4, stop on entering $[x, x+2]$. The stopped function is bounded on the traversed compact interval and the stopping time is finite, so the supermartingale inequality gives

$$G(x) \le \frac{11}{100}\exp\{(18/25)(x+6)\}, \qquad x \le -6.$$

For $K_2 \ge 3$, integrating this bound gives

$$t^-_{\rho^0}(K_2) \le \frac{11}{100}\,\frac{b^{2K_2-4} + b^{2K_2-5}}{1-b^2}, \qquad b = e^{-18/25}.$$

The rational right recursion and this geometric left bound are used in (SM4.1). All scalar comparisons precede the matrix stages.

SM3.2 Validated lower matrices and bootstrap

At saved endpoints, tail values are enclosed by the exact binary rationals plus or minus the strict full-field trajectory error. Across a cell, true tails are monotone in mean time. An atom bound over a cell of width h is $\max\{0, \widehat{a}_l - e_l\}\,\underline{e^{-2h}}$, where e_l bounds the left-endpoint atom error and the exponential is bounded below. Over candidate cycle k, nonexpansion and the period residual add at most kr to the discrepancy from the integer-translated first cycle.

The orbit is first enclosed analytically using the preceding tails. After a completed stage gives $E_{\mathrm{old}} \ge d_1(\nu, \rho^0)$, nonexpansion also encloses every orbit phase by first-cycle nodes with error $e_s + E_{\mathrm{old}}$, independently of cycle index. Intersect these valid bounds.

The midpoint-chain lower off-diagonals are the sums of the compared-law lower tails and atoms, because the rate twice a midpoint is that sum. Negative diagonals use validated upper exit bounds. After each period shift the finite frame by one and kill any path leaving it. For $A = -2I + C$, downward arithmetic computes

$$e^{hA} \;\ge\; \underline{e^{-2h}}\sum_{j=0}^{4}\frac{h^j C^j}{j!}$$

entrywise, with a nonnegative finite right side. Propagating all core columns and summing row minima supplies the stated common mass. The backward verifier uses its own validated bootstrap history. The final readout uses the larger forward/backward error bound; both bootstrap histories are included in the artifact.

■ SM4 Numerical certificate details

Let v be an exact rational probability profile of mean zero, obtained by normalizing the saved candidate and correcting its mean with an integer-shift mixture. Binary64 input values define exact rational data. Its strict one-period return error is $r < 4.601\,10^{-9}$. The full-cell trajectory error is e_s, including the mean correction; $e = \sup_s e_s < 4.585\,10^{-9}$. Bounds come from the full polynomial residual, including the top outflow. For a Bernstein residual of degree six, integration of its absolute value is bounded by the cell width times the sum of absolute coefficients divided by seven. Tail nonexpansion then controls the error against the true flow of v.

Proposition SM4.1 (A posteriori profile estimate)**.** Fix a core $|k| \le K_2$ on both channels. Suppose the exact midpoint difference kernel between the candidate flow and the orbit, over B mean periods with the co-moving shifts, has common mass at least $\beta > 0$ from every core starting state. For

$$t_p(K_2) = \sum_{\text{sides}} \{\mathbb{E}_p(X - K_2)_+ + \mathbb{E}_p(-X - K_2 - 1)_+\},$$

one has

$$d_1(v, \rho^0) \le Br/\beta + 2t_v(K_2) + 2t_{\rho^0}(K_2). \tag{SM4.1}$$

Proof. The signed tail difference has total mass zero. Its outside-core absolute mass is at most $t_v + t_{\rho^0}$. On the core the common kernel mass cancels at least β times the smaller of its positive and negative masses, reducing the ℓ_1 norm by at least $\beta[d_1 - 2(t_v + t_{\rho^0})]$. Iterating the one-period return bound and nonexpansion gives $d_1(v, T^B v) \le Br$. Triangle inequality between $v, T^B v, \rho^0 = T^B \rho^0$ now yields (SM4.1). □

The finite lower kernels retain only paths inside their declared moving frame, so deleting escaping paths preserves a lower bound. Validated exit upper bounds avoid unnecessary killing at high starting levels. Each lower Metzler matrix has form $-2I + C$, $C \ge 0$. Downward dyadic arithmetic, the nonnegative exponential terms through degree four, and a rational lower bound on e^{-2h} propagate all core columns. At each final row take the minimum over all core columns and sum these minima to obtain β. The backward implementation propagates all terminal indicators with 64-bit directed arithmetic; the forward computation uses 80 bits. Each bootstrap stage uses only an already completed earlier bound. Supplement section SM3.1 gives the analytic infinite-tail bounds required by (SM4.1).

The larger final forward/backward bound is the exact rational

$$E = \frac{1365032393294082761}{151115727451828646838272} < 9.034\,10^{-6}.$$

Nonexpansion and integer translation give uniform profile error $D_* = E + e < 9.038\,10^{-6}$ at every real mean.

For feedback, the origin coefficient and overflow, exact rational sums over 257 saved phase nodes (256 intervals) yield errors

$$\text{feedback: } 2D_* + \frac{9472}{180 \cdot 256^4}, \qquad \text{origin coefficient: } 8D_* + \frac{76288}{180 \cdot 256^4}, \qquad \text{overflow: } D_*.$$

The profile constants follow by coupling the two independent comparisons, a three-draw origin event, and positive part. For quadrature, the quadratic field has $\|F\|_1 = 2$, $\|DF\| \le 4$, $\|D^2F\| \le 8$ and higher field derivatives zero. Its trajectory derivative norms through order four are bounded by $2, 8, 64, 640$. Multinomial differentiation gives fourth derivative bounds 9472 for the feedback integrand and 76288 for the origin integrand. Composite Simpson quadrature has error $M_4/(180N^4)$; its positive weights do not amplify grid profile errors.

Table SM1 Preserved certificate stages. The first two stages miss the original 0.001 target. The fourth reaches the stricter 0.0001 stopping target. All distance bounds are rounded upwards.

Core radius	Periods	Frame radius	Subdivisions	Common mass	W_1 bound
4	8	8	64	0.0006913	0.050068
6	16	10	64	0.0651929	0.002797
8	24	12	128	0.5510914	0.000157
10	32	14	128	0.5811574	0.000009034

■ SM5 Statistical protocols and interpretation

All coverage statements below use the ideal word model: underlying words are uniform and independent within streams and across replicate labels. Fresh microscopic streams are also independent of prior history. SplitMix drivers and saved seed lists specify the reproducible implementation.

The original and prospective comparisons form separate inference families. Union bounds allow shared labels across cells. The prospective coverage statement averages over both the reused reference and fresh data. It retains the reference event's original error allocation.

Table SM2 Certified readouts of the limiting profile, with profile and quadrature errors included.

Readout	Certified interval
$-\overline{g}'(0)$	[0.71997987, 0.72012468]
$\overline{g}(1)$	[−0.62242623, −0.62239004]
$\overline{g}(2)$	[−0.90371675, −0.90368056]
$S_0(0)$	[0.63127654, 0.63129462]
$S_0(1)$	[0.75422265, 0.75424073]

■ SM5.1 Original allocation design

The model is exactly Section 2, with identity matchings in both directions or product crosses, $q \in \{64, 128, 256, 512\}$, $\lambda \in \{0, 1, 2\}$ and $\varepsilon = \lambda/q$. Capacity is the integer $c = q/2$, so $\kappa = 1$ and $n_* = q^2$. The respective replicate counts are 128, 128, 256, 1024. Every paired path runs through $q^2 + 2q$ arrivals. There are 4608 fixed/product pairs, or 9216 individual outcomes. The batch includes every planned label.

The 1024 seed labels are the first eight SHA256 bytes of the fixed string prefix and replicate number specified in the protocol, interpreted as a big-endian 64-bit word. Smaller scales use prefixes of this list. A counter-based SplitMix64 driver supplies two words per arrival; disjoint bit fields determine event type, direction and endpoints. All probabilities are exact power-of-two subdivisions at these parameter values. The same driver words are used within a fixed/product pair and across cells sharing a replicate label. Thus cells and checkpoints are dependent. The seed label is the inferential unit.

Compilation and deterministic one-step tests preceded the allocation batch. The model checks mass and capacity identities, endpoint counting, and left-tie handling. All 9216 full labelled critical configurations were subsequently re-read and independently checked against their mass, imbalance and overflow summaries. A second read-only audit reconstructed all 193536 saved checkpoint masses, overflows, product feedback counts and fixed joint-table marginals, then all 9216 outcome metrics; its largest metric roundoff difference was below $2.3\,10^{-16}$. At $\lambda = 0$ all 32256 paired recorded checkpoints agree, as required. All checkpoint side histograms, fixed joint tables, driver hashes and logs are retained.

For phase-aware feedback there are 16 prescribed checkpoints

$$n_j = q^2/2 + 2q\left\lfloor \frac{(q/4-1)j}{15} \right\rfloor + \frac{2qj}{16}, \quad j = 0, \dots, 15.$$

The common phase is exactly $j/16$. The feedback metric is the trajectory RMS of the exact conditional feedback minus the phase-aware response $g(h_n, \Delta_n/q)$. For overflow the metric is the maximum absolute error of $Z_c/q - S_h(\Delta/q)$ over the five fixed h values $-1, -1/2, 0, 1/2, 1$. The numerical lookup uses 257 old phase nodes with linear tail interpolation. Its uniform allowances are 0.00004861 for feedback and 0.00002440 for overflow, including $E + e$, interpolation and arithmetic. The interpolation addition in d_1 is $1/65536$, since the tail second-derivative norm is at most eight. Numerical allowances are used on both sides of each classification.

■ SM5.2 Original failure-rate and cross-kernel intervals

Failure-rate and cross-kernel CDF criteria are assessed at $q = 512$. Results at $q = 64, 128, 256$ describe the size dependence. Four feedback cells ($\lambda = 1, 2$ and both kernels) use RMS margin 0.10. Six overflow cells (all three λ and both kernels) use maximum-error margin 0.05. For each cell the target is a population probability of exceeding the margin at most 0.10. A numerically certain failure exceeds the margin even after subtracting its allowance; a possible failure exceeds it after adding the allowance. Exact binomial tail inversion on a conservative 0.001 grid bounds the failure probability below using certain failures and above using possible failures. Each side uses error probability $1/800$, giving simultaneous coverage at least 0.975 over the ten cells by union bound. Entirely interior intervals support the target, lower bounds above it detect mismatch, and all other cases are inconclusive.

Cross-kernel comparison uses endpoint CDFs of Δ_{q^2}/q and $Z_c(q^2)/q$ at $\lambda = 1, 2$. Common random numbers couple the simulation paths, whereas Section 3 uses maximal load-pair rank coupling. The prespecified practical CDF-distance margin is 0.20. DKW–Massart bounds [1] of width 0.057 for each of the eight empirical CDFs yield simultaneous pair bands 0.114, with family error at most 0.025 because

$$16\exp[-2 \cdot 1024 \cdot (0.057)^2] < 0.025.$$

The scalar inequality was checked with exact rational Taylor bounds. Observed KS distances, in order $\lambda = 1$ imbalance/overflow and $\lambda = 2$ imbalance/overflow, are $9/1024$, $17/1024$, $13/1024$ and $1/64$. The true CDF-distance upper bounds are respectively 0.122790, 0.130602, 0.126696 and 0.129625 after upward rounding. All are below 0.20, supporting this specified finite-size equivalence criterion. The resulting CDF-distance intervals have lower endpoint zero. Adding the primary and CDF family errors bounds the total simultaneous error by 0.05 under the reference model.

Table SM3 All primary $q = 512$ cells, with 1024 replicates per cell. All ten intervals support their prespecified population target. Intervals include the deterministic numerical allowance through certain/possible counts.

Metric	λ, kernel	Certain	Possible	Probability interval
Feedback RMS	1, fixed	0	0	$[0, 0.007]$
	1, product	0	0	$[0, 0.007]$
	2, fixed	0	0	$[0, 0.007]$
	2, product	0	0	$[0, 0.007]$
Overflow maximum	0, fixed	37	37	$[0.020, 0.058]$
	0, product	37	37	$[0.020, 0.058]$
	1, fixed	38	38	$[0.021, 0.059]$
	1, product	41	42	$[0.023, 0.064]$
	2, fixed	49	50	$[0.030, 0.073]$
	2, product	45	47	$[0.026, 0.070]$

■ SM5.3 Prospective design

The prospective protocol fixes $q = 512$, $\kappa = 1$, $\lambda = 1, 2$, matching/product kernels and 4096 shared replicate labels. Each path stops at q^2 arrivals. The source freeze is dated 17 October 2026, 23:42:34 UTC. The input freeze precedes the first trajectory, and the data freeze precedes inference.

For label i, form the string

`paper_micro_prospective_v1|replicate=%05d`

with i formatted as five digits. Interpret its first eight SHA256 bytes as a big-endian 64-bit seed. All 4096 values are unique and disjoint from the original and reference seed lists. The protocol fixes the sample size, seeds, and decision thresholds. The artifact contains the complete batch and its checks.

■ SM5.4 Prospective intervals and coverage allocation

The fixed-time nonnegative-product argument is a simple instance of betting-based mean inference [2]. For means, clip $V_i = \min(Z_i/q, 3)$ only for inference and retain all unclipped observations. Lower candidates $m \in [0, 1]$ use $\prod_i [1 + V_i - m]$; upper candidates $m \in [1/2, 3]$ use $\prod_i [1 + (2/5)(m - V_i)]$. Reject only at capital at least 1000, and invert on the fixed 10^{-5} grid outwards using exact integer products. Each endpoint costs at most 0.001 by the fixed-time nonnegative-product bound.

Changing one arrival changes final labelled-load ℓ_1 distance by at most two and overflow by at most one. At q^2 arrivals, bounded differences thus gives $\mathbb{P}(X - \mathbb{E}X > t) \le e^{-2t^2}$ for $X = Z/q$. The fixed coarse bound $U = \overline{X} + 0.030$ costs at most $e^{-2R(0.030)^2} < 0.001$ per cell. On $\mathbb{E}X \le U < 3$, add the upward-evaluated clipping-bias bound

$$\mathbb{E}(X - 3)_+ \le \frac{1}{4(3 - U)} e^{-2(3-U)^2}.$$

All four mean-width gates ≤ 0.04 passed. The mean intervals include the clipping-bias bound above.

The existing reference's four fixed mean endpoints each cost at most 0.001. Its clipped mean intervals are $[0.69875, 0.70333]$ at $\lambda = 1$ and $[0.67391, 0.67781]$ at $\lambda = 2$. The audited total approximation and truncation allowances 0.0006 and 0.0010 give true SDE mean intervals $[0.69815, 0.70393]$ and $[0.67291, 0.67881]$. Both satisfy the prespecified width gate 0.010. Subtraction of these intervals from the fresh unconditional micro intervals gives Table 1. Eight fresh betting endpoints, four fresh coarse bounds and four existing reference endpoints cost at most 0.016.

Fresh micro CDFs use DKW width $a = 0.032$ with $R = 4096$; the two existing reference CDF events use $b = 0.012$ with $N_{\mathrm{ref}} = 32768$. The audited reference-to-SDE coupling supplies horizontal radius $r_1 = 0.010$, $r_2 = 0.020$ and vertical allowance $\tau = 0.005$. With A the micro empirical law, B the computed reference empirical law, $B_\pm = B \pm r_\lambda$, $d(A, B) = \sup_x (F_A - F_B)$ and $\delta = a + b + \tau = 0.049$, the distance bounds are

$$L = \max\{0, \max[d(A, B_-), d(B_+, A)] - \delta\},$$
$$U = \min\{1, \max[d(A, B_+), d(B_-, A)] + \delta\}.$$

Every jump, including ties, is included; stored long-double hexadecimal values are parsed as exact rationals. The reference CDF-envelope width gate 0.10 passed for both λ values. The simultaneous failure allocation for this family's eight comparisons is

$$\begin{aligned}\alpha_{\text{mean}} &\le 0.016,\\ \alpha_{\text{CDF}} &\le 8e^{-2\cdot 4096(0.032)^2} + 4e^{-2\cdot 32768(0.012)^2},\\ \alpha_{\text{joint}} &< 0.018138367441 < 0.05.\end{aligned}$$

The scalar exponential bounds were certified before sampling with rational Taylor bounds. The error allocation gives at least 95% simultaneous coverage for the eight prospective comparisons.

■ SM5.5 Earlier post-hoc diagnostic

Before the prospective batch, the old $R = 1024$ micro endpoints had already been observed when an SDE follow-up recipe was designed. Its nominal micro mean intervals have no selection-adjusted coverage guarantee. The corresponding whole-family 95% statement was withdrawn. The corrected status records the remaining CDF-only anchor. See the preserved corrected diagnostic status.

■ SM6 SDE reference error budget

Write $b = \lambda\overline{g}$, $N = 16384$ and $h = 1/N$. The front representation gives $|b| \le \lambda$, $-2\lambda \le b' \le 0$ and $|b''| \le 24\lambda$. Under $2\lambda h \le 1$, the derivative satisfies $0 \le 1 + hb'(x) \le 1$, so $x + hb(x)$ is nondecreasing and 1-Lipschitz. The reference values $\lambda = 1, 2$ give $2\lambda h \le 1/4096$. Let $L = 2\lambda$ and $C_0 = 2\lambda^2 + 12\lambda$.

The local exact-versus-Euler remainder has the Itô representation

$$R_k = \int_0^h (h-u)b'(X_{kh+u})\,\mathrm{d}B_u + \int_0^h (h-u)(bb' + b''/2)(X_{kh+u})\,\mathrm{d}u.$$

Its conditional mean is bounded by $a = C_0h^2/2$. Its centered conditional L_2 norm is at most $s = Lh^{3/2}/\sqrt{3} + C_0h^2/2$. Contraction yields the second-moment recurrence

$$v_{k+1} \le v_k + 2a\sqrt{v_k} + a^2 + s^2.$$

Induction gives $\sqrt{v_N} \le \sqrt{N}s + Na$. Hence the Euler bound has an explicit constant and no exponential amplification.

The global feedback table uses Simpson quadrature over 256 phase intervals and linear interpolation at spacing $1/512$. Its error is at most

$$0.00004861 + \frac{9472}{180\cdot 256^4} + \frac{3}{512^2} < 0.000061.$$

For S_0, linear interpolation adds $1/(4\cdot 512^2)$ to 0.00002440, giving error below 0.000026. Certified analytic tails cover constant feedback extension beyond eight and the overflow extension $|d|/2$ beyond 32. All approximate feedback slopes lie in $[-2, 0]$, and all approximate overflow slopes lie in $[0, 0.5001]$.

For midpoint uniforms of cell width $\delta = 2^{-52}$, an explicit Box–Muller coupling gives

$$\mathbb{E}|Z - Z_{\text{mid}}|^2 \le 100\delta(1 + \log(1/\delta)) < 10^{-12}.$$

The implementation's series and rounding bounds are stated below. Minkowski across N scaled increments gives accumulated driver error at most $128(10^{-6} + 10^{-9}) < 0.00013$.

For clipping at three, the ideal-Gaussian Euler endpoint is a 1-Lipschitz function of its Gaussian increments. Oddness and contraction give its second moment at most one. The approximate readout has value below 0.632 at zero and Lipschitz constant 0.5001. Its mean is below 1.133. Gaussian concentration [3] gives clipping bias below

$$\frac{0.5001^2}{3 - 1.133}\exp\left[-\frac{(3-1.133)^2}{2(0.5001)^2}\right] < 0.00014.$$

The reference error allowances include the globally certified feedback/readout tables, Euler discretization, the Gaussian driver, state/readout arithmetic and truncation. The coefficient-table allowances are 0.000061 for $\overline{g}$ and 0.000026 for S_0, with checked readout Lipschitz bound 0.5001. The Gaussian-driver accumulated L_2 allowance is 0.00013. The arithmetic supplement bounds normal-coordinate arithmetic error below 10^{-12}, supporting the frozen 10^{-9} coordinate allowance; state rounding plus readout rounding divided by 0.5001 is below $4\,10^{-12}$, within the single shared 10^{-6} allowance. These statements are conditional on binary round-to-nearest arithmetic with at least 64 significand bits, normal finite operation ranges and exact radix-two exponent reduction. The historical reference has no recorded floating-point register snapshot; its arithmetic guarantee uses these stated assumptions. The audited version 2 arithmetic supplement is included with the source, together with the reference protocol and probability contract.

The reference mean budget uses two readout constants. The exact S_0 is $1/2$-Lipschitz, whereas the stored interpolator is 0.5001-Lipschitz. Let e_λ denote the certified Euler L_2 bound. With $N = 16384$, $L = 2\lambda$ and $C_0 = 2\lambda^2 + 12\lambda$, the monotone Euler-map estimate gives

$$e_\lambda = \frac{L/(17/10) + C_0/2}{N} + \frac{C_0}{2N\sqrt{N}},$$

using $\sqrt{3} > 17/10$ and $\sqrt{N} = 128$. For $\delta_g = 0.000061$ and $\delta_S = 0.000026$, the mean error is bounded by

$$\delta_S + \tfrac{1}{2}(e_\lambda + \lambda\delta_g) + 0.5001(0.00013 + 0.000001) + 0.00014.$$

It is less than 0.000514 at $\lambda = 1$ and 0.000857 at $\lambda = 2$. The resulting bounds fit the frozen allowances 0.0006 and 0.0010. The CDF coupling follows from

$$(e_\lambda + 0.00013)^2 \le 0.005(r_\lambda - \lambda\delta_g - 0.000001)^2.$$

The first two terms in the mean bound compare the true SDE to the ideal-Gaussian Euler process. The remaining terms cover the computed driver, joint arithmetic error and Gaussian clipping tail.

■ SM7 Reproducibility artifacts

The reproducibility repository is `https://github.com/19991111/critical-overflow-reproducibility`. Selected frozen inputs are supplied in this repository. Its file-level provenance map records their original commits, archive members, and checksums. The source commits below identify the original allocation batches, SDE reference, and numerical certificates. Section SM9 documents the sensitivity experiments supplied with the source archive.

The original allocation batch contains 9216 outcomes and 193536 checkpoint observations. Its original source commit is

`dddf58f43c0831e1c24e3017fc11618938453478`.

The prospective batch contains 16384 endpoints from 4096 shared labels. Its artifact map identifies source, protocol, seeds, configurations, hashes and logs. The original source commit is

`09480b6e9afcd54574245433df21c1cfac1eede6`.

The SDE reference originates from

`067543228ccdc4efbce5b8df7a0849b03f5fc1ec`.

Its reused status and original confidence events are part of the comparison design.

The accompanying `supporting_records_fcs.zip` contains the arithmetic supplement, probability contract, reference protocol, frozen prospective plan, and corrected post-hoc status. The same records are included in the `evidence/` directory of the LaTeX source bundle. Its `RECORDS_INDEX.txt` maps these files to the supplement sections. File manifests identify the input bytes.

The numerical audit recomputes all 256 full-cell residual bounds, 24 fluid readout estimates, 20482 coefficient nodes and 65536 reference endpoint readouts. It reduces the saved forward and backward terminal matrices using rational arithmetic. Matrix propagation is assessed through the comparison proof and archived implementation.

The endpoint audit recomputes mass, imbalance and overflow from the 25600 original and prospective allocation configurations. The statistical audit recomputes prospective betting endpoints and neighboring grids, clipping bounds, CDF suprema and family allocations. It also checks the original ten failure-rate intervals and four cross-kernel CDF comparisons. The original experiment artifact also records the full checkpoint reconstruction.

Recorded hashes and logs document the source and data chronology. Sections SM5 and SM6 specify the probability model and arithmetic assumptions used for the reported guarantees.

The LaTeX source bundle includes build instructions. The supporting records contain the numerical and statistical audit receipts.

■ SM8 Detailed contraction and attraction proof

We supply the analytic details behind main-article section 4. Fix $\theta = 1/200$. The lower bounds below depend on the initial laws and the finitely many states under consideration.

■ SM8.1 Flow, moments, and compact sets

The atom equations associated with the tail flow are

$$a_k' = 2a_{k-1}y_{k-1} - 2a_k y_k, \qquad b_k' = 2b_{k-1}x_k - 2b_k x_{k+1}. \tag{SM8.1}$$

The birth rates are at most two. On the affine tail space of finite-first-moment laws, the tail field is globally 4-Lipschitz. One can construct the flow by Picard iteration for the corresponding two inhomogeneous birth laws. Bounded rates give nonexplosion and preserve first moments on bounded time intervals. Uniqueness follows from the Lipschitz bound. In particular, the atoms and tails are continuous in time, and the mean clock satisfies $\mu' = 1$.

Here is a direct verification of the moment estimate used for compactness. Let $\nu_k = (a_k + b_k)/2$ and $w_k = a_k y_k + b_k x_{k+1}$. Then $\sum w_k = 1$, $w_k \le 2\nu_k$, and $\sum_{j\ge k} w_j = x_k y_k \le [(x_k + y_k)/2]^2$. Thus w is stochastically below the minimum of two independent draws from ν. For $Y = \theta(X - \mu)$, $X \sim \nu$, and an independent copy Y', this gives

$$\sum_k (w_k - \nu_k)\sinh(\theta(k-\mu)) \le -\tfrac{1}{2}\mathbb{E}|\sinh Y - \sinh Y'| \le -\tfrac{1}{2}(\mathbb{E}\cosh Y - 1).$$

The last inequality is proved in Supplement section SM1. Taylor's formula gives $\cosh(z+\theta) - \cosh z - \theta\sinh z \le (e^\theta - 1 - \theta)\cosh z$. Differentiating the moving-center moment, with $V = 2\mathbb{E}\cosh Y$, therefore yields

$$V' \le -\{\theta/2 - 2(e^\theta - 1 - \theta)\}V + \theta \le -\theta V/4 + \theta.$$

Bounded birth rates preserve a finite exponential moment on each bounded interval, justifying differentiation by domination. Hence $V \le M$ is invariant whenever $M \ge 4$.

For a fixed mean α, write $C_{\alpha,M} = \{z : \mu(z) = \alpha,\ V_\theta(z) \le M\}$. The moment sublevel $C_{\alpha,M}$ is compact in d_1. The exponential bound gives tightness and uniform integrability of first moments for each marginal, so every sequence has a Wasserstein-convergent subsequence. Means pass to the limit, and Fatou's lemma preserves the moment bound. The same argument proves compactness of

$$K_M = \{z : \mu(z) \in [0,1],\ V_\theta(z) \le M\}.$$

For $\alpha \in [0,1]$, identical mixtures of 0 and 1 with mean α have $V_\theta \le 2\cosh\theta < 4$. Thus all these slices are nonempty for $M \ge 4$.

■ SM8.2 Explicit positive lower bounds

Choose integers ℓ, r with $a_\ell(0) = A > 0$ and $b_r(0) = B > 0$. Variation of constants in (SM8.1) gives

$$\begin{aligned} a_k(t) &\ge a_k(0)e^{-2t} + 2\int_0^t e^{-2(t-s)}a_{k-1}(s)y_{k-1}(s)\,\mathrm{d}s, \\ b_k(t) &\ge b_k(0)e^{-2t} + 2\int_0^t e^{-2(t-s)}b_{k-1}(s)x_k(s)\,\mathrm{d}s. \end{aligned} \tag{SM8.2}$$

The initial atoms therefore retain Ae^{-2t} and Be^{-2t}. If $\ell \le r$, use $y_{j-1} \ge b_r$ to generate successively the left atoms at $\ell+1, \dots, r+1$. Then $x_{r+1} \ge a_{r+1}$ generates b_{r+1}. If $\ell > r$, use $x_j \ge a_\ell$ to generate the right atoms at $r+1, \dots, \ell$. In both cases, with $N = \max(\ell, r+1)$, both atoms at N are positive for every $t > 0$. From both atoms at k, first generate a_{k+1} using b_k, and then generate b_{k+1} using a_{k+1}. This proves positivity at every $k \ge N$ for all $t > 0$.

The induction gives explicit quantitative bounds. Suppose two factors in a birth flux have lower bounds $Ct^p e^{-\gamma t}$ and $Dt^q e^{-\eta t}$, with $C, D > 0$ and $p, q \ge 0$. Equation (SM8.2) bounds the generated atom below by

$$\frac{2CD}{p+q+1}\,t^{p+q+1}e^{-(2+\gamma+\eta)t}. \tag{SM8.3}$$

Starting with Ae^{-2t} and Be^{-2t} yields explicit positive constants at every finitely generated level. On $[a,b] \subset (0,\infty)$ such a bound is at least $Ca^p e^{-\gamma b} > 0$. A high atom also bounds each lower finite tail from below.

■ SM8.3 A common state and strict cancellation

For two flows, set $d_k = x_k - \widetilde{x}_k$, $e_k = y_k - \widetilde{y}_k$ and use bars for arithmetic midpoints. Bilinear expansion gives exactly

$$\begin{aligned} d_k' &= 2\bar{y}_{k-1}(d_{k-1} - d_k) + 2\bar{a}_{k-1}e_{k-1}, \\ e_k' &= 2\bar{x}_k(e_{k-1} - e_k) + 2\bar{b}_{k-1}d_k. \end{aligned}$$

This is the forward equation of the main article's midpoint chain. Its total exit rates from X_k, Y_k are $2\bar{y}_{k-1}, 2\bar{x}_k \le 2$. Its transition kernel $P_{0,t}(i,j)$ is positive and mass preserving. Bounded rates justify its ordered-jump expansion on the countable state space and the resulting nonexpansion on signed ℓ_1 masses.

Fix $t > 0$ and two finite starting states. Choose $0 < a < b < t$. The previous subsection makes all same-channel forward rates positive on $[a,b]$ and all crossing rates positive at sufficiently high levels. Choose K above both starting levels with $K-1$ in that positive region. An X state can move forward to X_K; a Y state can move forward to Y_{K-1} and then cross to X_K. If one such path has L edges with rate lower bounds $c_1, \dots, c_L > 0$ on $[a,b]$, its ordered-jump integral gives

$$P_{0,t}(i, X_K) \ge e^{-2t} \frac{(b-a)^L}{L!} \prod_{j=1}^{L} c_j > 0. \tag{SM8.4}$$

The exponential accounts for no unlisted jumps before, between, or after the prescribed jumps. For $L = 0$, use the no-jump probability e^{-2t}. The rate bounds are supplied by (SM8.3), including the midpoint factor $1/2$ if only one of the two flows is used.

Let $v = (d,e)$ initially. Equal means imply $\sum_i v_i = 2(\mu - \widetilde{\mu}) = 0$. If $v \neq 0$, choose i, j with $v_i > 0$ and $v_j < 0$. Write $r = v^+ P_{0,t}$ and $s = v^- P_{0,t}$. Positivity and mass preservation give

$$\begin{aligned} \|vP_{0,t}\|_1 &= \|v\|_1 - 2\sum_k \min(r_k, s_k) \\ &\le \|v\|_1 - 2\min\{v_i P_{0,t}(i, X_K), (-v_j) P_{0,t}(j, X_K)\} < \|v\|_1. \end{aligned} \tag{SM8.5}$$

This proves strict distance decrease for every positive time and every distinct equal-mean pair with finite first moments.

■ SM8.4 Fixed points and uniform attraction

For $M \ge 4$, the continuous map $T = \mathsf{S}_{-1}\Phi_1$ preserves $C_{\alpha,M}$. It is nonexpansive and strictly shortens distances between distinct points. Compactness then gives a unique attracting fixed point. For $z \in C_{\alpha,M}$ let $a_n = d_1(T^n z, T^{n+1} z) \downarrow r$. Take a convergent subsequence $T^{n_j} z \to w$. Continuity gives both $d_1(w, Tw) = r$ and $d_1(Tw, T^2 w) = r$. Strictness forces $Tw = w$, and hence $r = 0$. Two distinct fixed points contradict strictness, so the fixed point is unique. For any initial z, let $d_1(T^n z, w) \downarrow l$. A convergent subsequence with limit v gives $d_1(v,w) = d_1(Tv, w) = l$. If $v \neq w$, strictness contradicts equality. Thus $l = 0$ and every orbit converges to w.

Take the fixed point z_* in $C_{0,4}$ and set $\rho^u = \mathsf{S}_{\lfloor u \rfloor}\Phi_{\{u\}} z_*$. The identity $\Phi_1 z_* = \mathsf{S}_1 z_*$ and commutation with integer shifts give continuity across integer phases, $\rho^{u+1} = \mathsf{S}_1 \rho^u$, and $\Phi_t \rho^u = \rho^{u+t}$. The moment bound gives $V_\theta(\rho^u) \le 4$. Each ρ^α is the fixed point in every $C_{\alpha,M}$, $M \ge 4$. Every finite-first-moment periodic family has a fixed point on each mean slice. Strict contraction identifies that point with ρ^α.

For $z \in K_M$, put

$$f_n(z) = d_1(T^n z, \rho^{\mu(z)}).$$

The mean map and the family ρ^u are continuous, so each f_n is continuous. Nonexpansion and $T\rho^{\mu(z)} = \rho^{\mu(z)}$ give $f_{n+1} \le f_n$. The preceding slice argument gives $f_n(z) \to 0$ for every $z \in K_M$. For $\delta > 0$ the relatively open sets $U_n = \{z \in K_M : f_n(z) < \delta\}$ are increasing and cover K_M. Compactness yields a finite subcover. Taking its largest index N gives $K_M = U_N$, and therefore $\sup_{z \in K_M} f_n(z) \le \delta$ for every $n \ge N$. For $t = n + s$, $0 \le s < 1$, shift invariance and nonexpansion imply

$$d_1(\Phi_t z, \rho^{\mu(z)+t}) = d_1(\Phi_s T^n z, \Phi_s \rho^{\mu(z)}) \le f_n(z).$$

Integer shifts reduce arbitrary initial means to $[0,1]$.

Finally, all orbit atoms are positive. Starting from any phase u, the positivity argument gives a level N above which atoms are positive at every later time. For an arbitrary k, choose an integer $n > 0$ with $k + n \ge N$. Then the atom at $k+n$ in ρ^{u+n} is positive, and the period relation identifies it with the atom at k in ρ^u. Each marginal increases stochastically along the birth flow. For $u \le v$, its Wasserstein distance is its mean increment. Summing the two increments gives $d_1(\rho^u, \rho^v) = 2(v-u)$, completing the profile theorem.

■ SM9 Sensitivity to tie rules and internal graphs

We vary the tie rule and internal candidate graph to test persistence and amplification of internal overflow. The source and protocol were committed before the native Linux run. The source commit was `3daccb3bdb4503a79accc26ba2f214bf9d79be30`. The source archive accompanying the manuscript contains a standalone reproduction directory named `reproducibility/mechanism_robustness_v1/`.

SM9.1 Experimental design and sampling

Every cell starts empty and has $q = 128, 256$, or 512 buckets per community, capacity $c = q/2$, and $n_* = q^2$ arrivals. The cross probability is $1/q$, with product cross sampling. Thus $\kappa = \lambda = 1$. One independent event word selects a cross with probability exactly $1/q$ and selects the community or cross direction with probability $1/2$. Two further words supply the left index and right selector. The power-of-two sizes permit exact uniform choices by masking. An independent stream breaks ties fairly in the fair-tie cells.

Each cell uses one of four fixed internal candidate laws. Complete sampling chooses both endpoints independently and uniformly. The permutation law samples a uniform left index and one of 16 fixed permutations uniformly, then uses the corresponding right image. Each community has its own permutations.

Repeated edges retain their multiplicity, giving a regular bipartite multigraph. Four independently generated graph pairs are used at each size. The cycle law samples $j = i$ or $j = i + 1$ modulo $q/2$ with equal probability. The matching law uses $j = i$ and serves as a disconnected control. All four laws have uniform endpoint marginals.

There are 1024 trajectories in each of the 24 cells. For the permutation law, four fixed graph pairs each supply 256 trajectories. At each size, configurations share 1024 arrival-seed labels. The tie streams and graph streams are separate. Python's operating-system random source generates the saved 64-bit seeds once, before simulation; the allocation streams use `mt19937_64`. The fixed graphs use Fisher–Yates permutations with rejection sampling.

SM9.2 An analytic matching control

For a separate analytic control, take no cross arrivals, so $\lambda = 0$, and use a fixed matching internally in each community. Let $c = q/2$ and $n_* = q^2$. There are q matching pairs across the two communities. Each arrival chooses one pair uniformly. Under every greedy tie rule, a pair with X assigned items has sorted loads $\lfloor X/2 \rfloor, \lceil X/2 \rceil$. Its overflow at capacity $q/2$ is exactly $(X - q)_+$. For one pair, $X \sim \mathrm{Bin}(q^2, 1/q)$, hence

$$\mathbb{E}Z_c(q^2) = q\mathbb{E}(X - q)_+.$$

The binomial central limit theorem gives $(X - q)/\sqrt{q} \Rightarrow N(0, 1)$. Its second moments equal $1 - 1/q$, so uniform integrability passes the positive-part expectation to $(2\pi)^{-1/2}$. Consequently

$$\mathbb{E}Z_c(q^2) = \{(2\pi)^{-1/2} + o(1)\}q^{3/2}.$$

Community choices remain independent fair choices, so $\mathbb{E}\Delta^2 = q^2$ and $\mathbb{E}|\Delta|/2 \le q/2$. Subtracting this community-excess term gives the same leading asymptotic for $\mathbb{E}G_c$. This internal matching law has uniform endpoint marginals but order-$q^{3/2}$ overflow, showing that internal marginals alone do not determine the overflow scale.

- **Weak cross sampling.**

Retain the fixed internal matchings and use the product cross law of Section SM9 with cross probability ε_q. Start both coupled processes empty, with $c = q/2$ and $n_* = q^2$. Share every internal arrival's community and matching edge. At each cross arrival, apply the original cross update in one process. Give the zero-cross process a fresh uniform choice among all q internal matching pairs. Each reference arrival is therefore uniform on these pairs, conditional on the joint past and the event type, giving the zero-cross law above.

Use left-favored ties in both processes, or share one fresh independent fair-tie coin at every arrival. Greedy integer updates with the same candidates and tie priority are nonexpansive in labelled ℓ_1 distance. Write $d = a - b$ for two integer load vectors. If the updates select i in a and $j \ne i$ in b, shared tie priority gives $d_i < d_j$. The updated discrepancies are $d_i + 1, d_j - 1$, whose absolute sum cannot increase. At a replaced event, first compare the states after the same cross update. Changing that pair to an internal pair in the reference costs at most two more in ℓ_1, since both updates add one unit to the same state. If $N_\times(n)$ counts replaced arrivals, induction gives

$$\|A^\varepsilon(n) - A^0(n)\|_1 \le 2N_\times(n). \tag{SM9.1}$$

At n_* the load vectors have equal total q^2. The difference of the load vectors has equal positive and negative masses. Since $(x - c)_+$ is nondecreasing and 1-Lipschitz, their total overflow differs by at most half their ℓ_1 distance. Writing $N_\times = N_\times(n_*)$ gives

$$\begin{aligned} |Z_c^\varepsilon - Z_c^0| &\le \tfrac{1}{2}\|A^\varepsilon - A^0\|_1 \le N_\times, \\ |\Delta^\varepsilon - \Delta^0| &\le \|A^\varepsilon - A^0\|_1 \le 2N_\times. \end{aligned} \tag{SM9.2}$$

The identity $G_c = Z_c - |\Delta|/2$ at critical fill then yields

$$|G_c^\varepsilon - G_c^0| \le |Z_c^\varepsilon - Z_c^0| + \tfrac{1}{2}|\Delta^\varepsilon - \Delta^0| \le 2N_\times. \tag{SM9.3}$$

For $\varepsilon_q q \to \lambda \in [0, \infty)$, $\mathbb{E}N_\times = \varepsilon_q q^2 = O(q)$. Combining this bound with the zero-cross calculation gives

$$\mathbb{E}G_c^\varepsilon(q^2) = \mathbb{E}G_c^0(q^2) + O(q) = \{(2\pi)^{-1/2} + o(1)\}q^{3/2}. \tag{SM9.4}$$

Equation (SM9.2) gives the same leading term for $\mathbb{E}Z_c^\varepsilon(q^2)$. The case $\varepsilon_q = 1/q$ gives the sensitivity experiment's $\lambda = 1$ regime.

Table SM4 Differences in mean omitted overflow G_c/q at $q = 512$, $\kappa = \lambda = 1$. Each contrast uses 1024 paired labels. The intervals belong to the simultaneous 95% family of 60 quantities defined above.

Tie rule	Internal graph minus complete	Difference	Interval
Left	Cycle	1.9104	[1.6787, 2.1421]
Left	Matching	8.0681	[7.8364, 8.2997]
Fair	Cycle	1.9489	[1.7173, 2.1806]
Fair	Matching	8.0440	[7.8123, 8.2757]

■ SM9.3 Readouts and uncertainty summaries

The primary readouts are Z_c/q, $|\Delta|/(2q)$, and their difference G_c/q at critical fill. Table SM5 gives every cell's mean omitted contribution. Monte Carlo standard error (MCSE) is sample standard deviation divided by the square root of the number of trajectories. For the equally weighted permutation-graph mean, it is

$$\mathrm{MCSE} = \frac{1}{4}\left(\sum_{b=1}^{4} \frac{s_b^2}{256}\right)^{1/2},$$

where s_b^2 is the sample variance within fixed graph pair b. The MCSE conditions on these four graph pairs. Graph-specific summaries in the source archive report variation between them. The paired fair-minus-left difference uses the same arrival labels. For permutation graphs, the same stratified formula applies to the paired differences within each graph pair.

The last column of Table SM5 uses the original frozen left-tie complete-graph readout $S_0(\Delta/q)$ without recalibration. It is a transfer diagnostic for the changed dynamics. The secondary window $|\Delta|/q \le 1/8$ was also fixed before the run. Table SM6 gives counts and conditional overflow means for this window. Section 6.5 distinguishes the window and profile readouts.

■ SM9.4 Simultaneous intervals for means and topology contrasts

We supplement the original MCSE summaries with an analysis of the saved trajectories. The family contains all 24 cell expectations and all 36 internal-graph contrasts within a fixed size and tie rule. The latter count is $3 \cdot 2 \cdot \binom{4}{2}$. All expectations condition on the fixed internal graphs.

Within each cell, use the independent per-arrival draws and independent trajectory labels of the allocation model. Changing one input changes at most one allocation at that arrival. Subsequent greedy integer updates with the same candidates and tie draws are nonexpansive in labelled ℓ_1 distance. For different winners, the corresponding integer discrepancies $d_i < d_j$ become $d_i + 1, d_j - 1$, so their absolute sum cannot increase. Thus the two endpoint load vectors A, A' have equal totals and satisfy

$$\|A - A'\|_1 \le 2, \qquad |Z_c(A) - Z_c(A')| \le 1, \qquad \big||\Delta(A)|/2 - |\Delta(A')|/2\big| \le 1.$$

Consequently one input changes G_c/q by at most $2/q$.

Write $N = 1024$, and let $\widehat{\mu}$ denote a cell mean with expectation μ. There are Nq^2 independent inputs; changing one affects this mean by at most $2/(qN)$. The bounded-difference inequality [3, Theorem 3.11] gives

$$\mathbb{P}\{|\widehat{\mu} - \mu| \ge u\} \le 2e^{-Nu^2/2}. \tag{SM9.5}$$

The argument also covers the four fixed permutation-graph strata, since all 1024 trajectories have equal weight.

For $K = 24$ cell means and $\alpha = 0.05$, set

$$a = \sqrt{\frac{2}{N}\log\frac{2K}{\alpha}} < 0.1159, \qquad b = 2a < 0.2317. \tag{SM9.6}$$

Equation (SM9.5) bounds each two-sided failure probability by α/K. A union bound places all 24 expectations in $[\widehat{\mu} - a, \widehat{\mu} + a]$ simultaneously with probability at least 0.95.

On the same event, every graph contrast $\delta = \mu_g - \mu_{g'}$ satisfies

$$|\widehat{\delta} - \delta| \le |\widehat{\mu}_g - \mu_g| + |\widehat{\mu}_{g'} - \mu_{g'}| \le b.$$

Thus the 24 mean intervals and 36 contrast intervals have joint coverage at least 95%, with family-wise noncoverage at most 0.05. This argument allows arbitrary dependence between cells, including their shared arrival labels. All 24 mean intervals have positive lower endpoints.

Table SM4 gives the four comparisons highlighted at $q = 512$. Interval endpoints are rounded outwards to four decimal places. The source archive includes `sensitivity_inference.py` and `verification/sensitivity_intervals_v1/`, with all 60 intervals and the hashes of their fixed inputs.

Table SM5 Every sensitivity cell: mean omitted overflow, MCSE, and mean absolute error (MAE) of the original frozen profile readout. Each row contains 1024 endpoints; $\kappa = \lambda = 1$ throughout.

q	Internal graph	Tie rule	Mean G_c/q	MCSE	Profile MAE
128	Complete	left	0.38638	0.00483	0.03530
128	Complete	fair	0.41048	0.00507	0.04006
128	16 permutations	left	0.41046	0.00489	0.03931
128	16 permutations	fair	0.43465	0.00514	0.05613
128	Cycle	left	1.44048	0.00957	1.07252
128	Cycle	fair	1.48312	0.00967	1.11714
128	Matching	left	3.94583	0.01240	3.58960
128	Matching	fair	3.94759	0.01241	3.59106
256	Complete	left	0.39887	0.00449	0.02433
256	Complete	fair	0.42260	0.00472	0.03345
256	16 permutations	left	0.42425	0.00468	0.03277
256	16 permutations	fair	0.45182	0.00484	0.05828
256	Cycle	left	1.83757	0.00965	1.46833
256	Cycle	fair	1.89215	0.00969	1.52282
256	Matching	left	5.82412	0.01275	5.46503
256	Matching	fair	5.82507	0.01277	5.46571
512	Complete	left	0.39581	0.00454	0.01735
512	Complete	fair	0.42022	0.00473	0.03225
512	16 permutations	left	0.41941	0.00468	0.02822
512	16 permutations	fair	0.44624	0.00484	0.05763
512	Cycle	left	2.30620	0.00993	1.94333
512	Cycle	fair	2.36916	0.01001	2.00659
512	Matching	left	8.46387	0.01320	8.11062
512	Matching	fair	8.46421	0.01320	8.11094

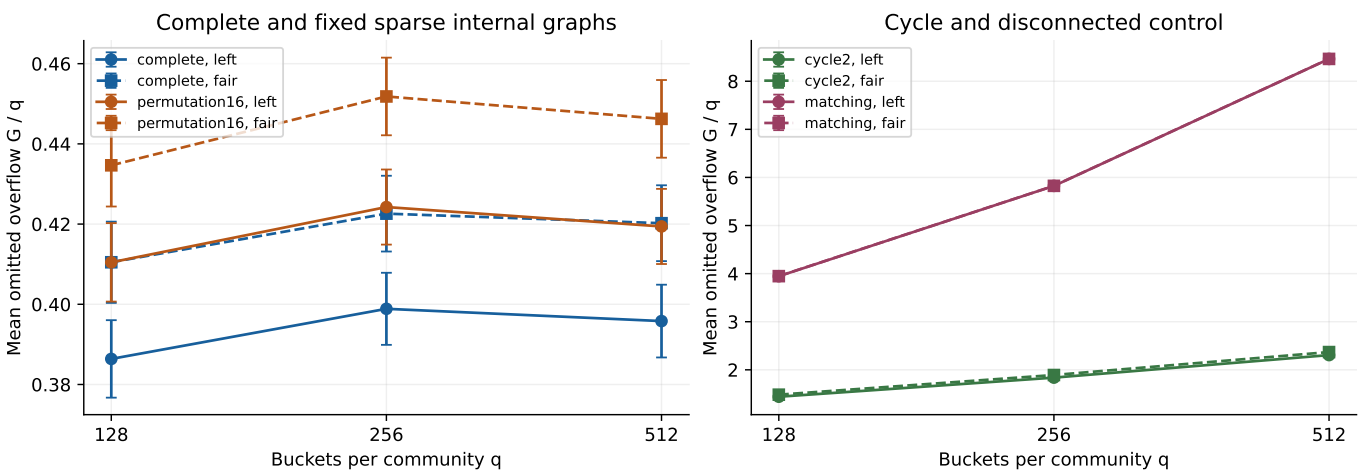


Fig. SM1 The omitted internal contribution under changed tie rules and internal graphs. Bars show twice MCSE conditional on the fixed graphs. The two panels use different vertical scales. Both tie rules show growing normalized contributions for cycles and matchings across all three tested sizes.

Table SM6 Secondary, prespecified near-balance window at $q = 512$: $|\Delta|/q \le 1/8$. Counts are out of 1024 per cell. The final column is the conditional mean of Z_c/q.

Internal graph	Tie rule	Count	Mean overflow
Complete	left	151	0.63380
Complete	fair	133	0.66671
16 permutations	left	142	0.65677
16 permutations	fair	136	0.69182
Cycle	left	112	2.63864
Cycle	fair	119	2.71307
Matching	left	116	8.88482
Matching	fair	113	8.89045

SM9.5 Execution and verification records

One hosted Ubuntu 24.04 job compiled the frozen C++ source with GCC 13.3.0 and executed the complete experiment on native Linux. Seeds, source hashes, compiler information, raw endpoint tables and analysis outputs were preserved before interpretation. All 24576 endpoints pass mass conservation and the exact integer identity in equation 1.1. In addition, 384 left-tie trajectories check the relabeling in Corollary 2.6 at every arrival and at the full endpoint. The mirrored process favors right ties and swaps candidate side labels and cross directions. The 384 mirrored checks reuse the left-tie trajectories.

A separate saved-data audit verifies all 135 returned file hashes, the source correspondence, and every endpoint identity. It recomputes the reported means and MCSEs from raw integer counts and checks cell completeness and label uniqueness. The archive includes the fixed protocol, source, seeds, raw tables, environment, graph-specific results, paired differences and audit receipt.

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